

Prabhaav Pillai

Computer Science Graduate Student & Software Engineer

+17047135246 | pillaiprabhaav@gmail.com | prabhaav.me | linkedin.com/in/prabhaavp | github.com/prabhaavp

EDUCATION

Michigan State University

MS in Computer Science

East Lansing, MI

Jan. 2026 – Dec. 2026

- GPA: 4.00/4.00
- Currently pursuing M.S. Accelerated Computer Science Program (Completed 9 graduate credits during B.S. via MSU Shared Enrollment program)

Michigan State University

BS in Computer Science

East Lansing, MI

Aug. 2023 – Dec. 2025

- GPA: 3.89/4.00 | Honors College | Minor in Mathematics
- Completed 4-year B.S. degree in 2.5 years

EXPERIENCE

AI Document Intelligence Extern (Pfizer Program)

Extern

July 2026 – Present

Remote

- Building an AI document intelligence system to automate data extraction, classification, and compliance risk detection across pharmaceutical records.
- Developing Python pipelines utilizing Layout-Aware OCR and RAG to query and extract structured insights from complex vendor files.
- Implementing document classification, dynamic file routing, and open-source LLM workflows integrated with a Gradio chatbot interface.

Software Engineering Capstone Developer

United Wholesale Mortgage

Aug. 2025 – Dec. 2025

Pontiac, MI

- Developed full-stack IT Goals Dashboard using React and ASP.NET Core to visualize real-time KPIs for 2,000+ IT professionals.
- Built EF Core migrations, backend metric aggregation logic, dynamic gauge widgets, and sparkline UI visuals exceeding SonarQube quality standards.
- Collaborated with product owners to refine yearly/monthly metrics, improving operational transparency across IT teams.

Volunteer CS Teaching Assistant

Microsoft TEALS Program

Aug. 2023 – June 2025

Detroit, MI

- Instructed 50+ students across two AP Computer Science classrooms in computational thinking, JavaScript, and algorithmic problem-solving.
- Coordinated with lead teachers to optimize lesson delivery, facilitate hands-on coding labs, and assist high school seniors with college readiness.

PROJECTS

Automotive Edge AI System - StarPilot Developer

July 2026 – Present

- Appointed to core dev team with repo write access; authored and merged **+3.5k lines of C++/Python/GLSL** to deploy real-time blind-spot monitoring and a custom **GLSL shader** Picture-in-Picture (PiP) side camera preview system.
- Trained custom **YOLO26n ONNX** models on 27+ hours of crowdsourced driving video using semi-automated CAN-bus radar logging for ground truth.
- Engineered real-time edge performance on **Snapdragon SoCs** using zero-copy NV12 buffer slicing and CPU core isolation under 1.2x throttle overhead.
- Built **Flask REST APIs** and **Galaxy web tools** for feature configuration, driving widespread adoption (**36K+ views, 180+ upvotes**) and maintaining production bug fixes.

CERTIFICATIONS

- Certified Cloud Practitioner (Score: 981/1000) - Amazon Web Services (July 2026)

SELECTED HONORS

- First Place Award, Education Category - MSU University Undergraduate Research Forum (UURAF, April 2024)
- Researched academic impacts of Large Language Models (LLMs) on student performance using mixed quantitative and qualitative analysis.

TECHNICAL SKILLS

Languages: Python, C++, C#, JavaScript, Java, SQL, GLSL

AI / Machine Learning: PyTorch, YOLOv8/26, Computer Vision, RAG, Layout OCR, ONNX Runtime, Model Optimization

Edge AI & Systems: Embedded AI, Real-time Inference, OpenGL, Linux, CMake, Performance Profiling, Networking

Frameworks & Tools: React, ASP.NET Core, Docker, REST APIs, Git, GitHub, Swagger UI, Fast API, Jupyter Notebook

Cloud & Databases: AWS (Certified Cloud Practitioner), Azure SQL, Entity Framework Core